

Webvoltz Private Limited

Technical Team Lead (MERN) Practical Assessment

Overview

You are provided with a partially developed collaboration tool called **CollabHub**.

The application combines:

- Real-time chat (via WebSockets)
- Channel-based collaboration
- Task management within channels

The current codebase contains:

- Bugs
- Incomplete features
- Structural and coding issues

Your role is to approach this as a **Technical Team Lead** and evaluate, debug, and plan improvements under a tight deadline.

Objectives

We are evaluating your ability to:

- Understand an existing codebase
 - Identify issues and suggest improvements
 - Debug real-world problems
 - Think from a Tech Team Lead perspective (planning, prioritization, decision-making)
 - Communicate your thoughts clearly
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What You'll Get

- [GitHub repository](#) (Full-stack MERN app)
- Project SOW - Attached with email

Tasks

1. Codebase Review

Review the existing codebase and provide:

- Top **3 critical issues** in the codebase
- Suggested improvements for each issue

Focus on:

- Architecture
 - Code quality
 - Scalability
 - Maintainability
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2. Bug Fixing

Known Issues:

1. Chat messages sometimes appear multiple times
2. Task creation/status updates are not reflected properly in UI

Your Task:

- Identify the **root cause**
 - Fix these issues
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Hidden Issue:

- Identify **at least 1 additional bug** not listed above - No need to fix that one
 - Explain:
 - Root cause
 - Fix approach
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3. Refactoring Approach

You are NOT required to refactor the entire project.

Instead, provide a plan:

- What architectural issues exist in the project?
 - How would you restructure the project?
 - What architectural improvements would you introduce?
 - What would you prioritize first?
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4. Execution Plan of Partial Implementation

As we've incomplete features in place. You got the features list (Scope of work) documentation. Based on that and compare it to the current situation, I need to provide high-level estimation and work plan.

Assume:

- Tight deadline (Need to complete as early as possible)
- Team of 2 developers

Provide:

- Work plan
 - Prioritization
 - Timeline (high-level)
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6. Communication Task

Record a **2–3 minute video** explaining:

- Your understanding of the system
- Key issues you identified
- How you approached solving them
- How you would lead the team moving forward

This is to evaluate:

- Communication clarity
- English proficiency
- Leadership thinking

Submission Requirements

Please submit:

1. Updated GitHub repository
2. README including:
 - Summary of changes for each task/bug
3. Short document (or README section) covering:
 - Code review points
 - Bug explanations brief
 - Refactoring approach
 - Execution plan with high-level timeline
4. Video recording (2–3 mins)

Expected Time

- Approx. **3-4 hours**

Notes

- Focus on **clarity of thinking**, not perfection
- We value **problem solving, clear understanding, decision-making and trade-offs**
- Keep solutions practical and realistic

Good luck! 